Coinciding symmetries determine a homogeneous mean

Principia Math In Progress

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Abstract

We prove Hypothesis 1 in the final version of Mihoković's *Coinciding Mean of the Two Symmetries on the Set of Mean Functions*. In fact the two symmetry operations, together with only a second-order diagonal expansion, determine the entire symmetric homogeneous mean. The result applies to every real parameter, as required by the final article, and does not assume differentiability away from the diagonal.

Task 1: the current source question

A strict symmetric mean is a function $M : (0,\infty)^2 \to (0,\infty)$ with $M(a,a) = a$, $M(a,b) = M(b,a)$ and $a < M(a,b) < b$ whenever $a < b$. It is homogeneous when $M(\lambda a, \lambda b) = \lambda M(a,b)$ for $\lambda > 0$. For another strict symmetric mean N, the group symmetry in [1, Definition 1 and (4)] is, off the diagonal,

$$S_M(N)(a,b) = \frac{a(u-a)(m-b)^2 - b(m-a)^2(u-b)}{(u-a)(m-b)^2 - (m-a)^2(u-b)}, \quad m = M(a,b), \quad u = N(a,b). \tag{1}$$

On the diagonal its value is a. The functional symmetry is characterized by

$$M\big(N(a,b), \sigma_M(N)(a,b)\big) = M(a,b). \tag{2}$$

The question concerns means for which these symmetries coincide for every N. Hypothesis 1 on page 15 of the final article asks whether the asymptotic coefficients of every such homogeneous mean agree with those of L_c for some $c \in \mathbb{R}$.

The MathDB wording and arXiv manuscript restrict c to $[-1,\infty)$. The final article corrects this to $\mathbb{R}$: its Corollary 1 and Theorem 3 also use all real c. We solve this current formulation. The old restricted-range statement is already superseded and is not the basis for new credit here.

Theorem 1. *Suppose M is a strict symmetric homogeneous mean, the identity (2) holds with $\sigma_M(N) = S_M(N)$ for every strict symmetric mean N, and there are real β, c such that*

$$M(x-1, x+1) = x + \beta + \frac{c}{x} + o(x^{-1}) \qquad (x \to \infty). \tag{3}$$

Then $\beta = 0$ and $M = L_c$ everywhere, where

$$L_c(a,b) = \begin{cases} \dfrac{(1+2c)(a+b) + \sqrt{(a+b)^2 + 4c(1+c)(b-a)^2}}{4(1+c)}, & c \neq -1, \\ \dfrac{2ab}{a+b}, & c = -1. \end{cases} \tag{4}$$

Conversely, for every real c, L_c is a strict symmetric homogeneous mean and its two symmetries coincide for every N.

The source's full asymptotic expansion implies (3). Thus the theorem proves more than coefficient agreement, under weaker expansion requirements. The possible constant β is included to avoid assuming in advance that symmetry alone excludes an absolute-value term at the diagonal.

The local perturbation

Fix $0 < a < b$, let $m = M(a, b)$, and write $d = m - a > 0$, $e = b - m > 0$. For $u = m + h$ let v denote the right side of (1). Its denominator is positive for $a < u < b$: it is $(u - a)e^2 + d^2(b - u)$. Direct simplification gives

$$v = m - \frac{h}{1 + \alpha h}, \qquad \alpha = \frac{e - d}{de} = \frac{a + b - 2m}{(m - a)(b - m)}. \tag{5}$$

In particular, as $h \to 0$,

$$v = m - h + \alpha h^2 + O(h^3), \qquad s := \frac{u + v}{2} = m + \frac{\alpha}{2}h^2 + O(h^3), \qquad r := \frac{|u - v|}{2} = |h| + O(h^2).$$

These perturbations are realized by admissible means. Choose $R = A$, the arithmetic mean, if $A(a, b) \neq m$; otherwise choose the harmonic mean $R = H$, since $H(a, b) < A(a, b)$ off the diagonal. For $0 < \lambda < 1$, the mean $N_\lambda = (1-\lambda)M+\lambda R$ is strict, symmetric and homogeneous, and $h = \lambda(R(a, b)-m)$ is nonzero and tends to zero from one side. The symmetry hypothesis therefore gives $M(u, v) = m$ along this limit. If the tested means are also required to possess full asymptotic expansions, these convex combinations do whenever M does.

By homogeneity, (3) is equivalent to

$$M(1 - z, 1 + z) = 1 + \beta z + cz^2 + o(z^2) \qquad (z \downarrow 0).$$

Since M is symmetric, we may apply it with $z = r/s \geq 0$ and obtain

$$m = M(u, v) = s + \beta r + c\frac{r^2}{s} + o(r^2/s). \tag{6}$$

After subtracting m and dividing by $|h|$, the limit in (6) gives $\beta = 0$. Dividing next by h^2 yields

$$\frac{\alpha}{2} + \frac{c}{m} = 0.$$

Substituting (5) and multiplying by the nonzero quantity $m(m - a)(b - m)$ shows that

$$Q_c(a, b; m) := 2(1 + c)m^2 - (1 + 2c)(a + b)m + 2cab = 0. \tag{7}$$

All limits were one-sided along admissible means; no differentiation of M at (a, b) was used.

The unique interior root and the converse

For every real c,

$$Q_c(a, b; a) = a(a - b) < 0, \qquad Q_c(a, b; b) = b(b - a) > 0.$$

A polynomial of degree at most two with these endpoint signs has exactly one root in (a, b). Indeed, existence follows by continuity; two distinct interior roots would force the endpoint signs to agree, and a double root does not change sign. For $c = -1$, the polynomial is linear and the root is $2ab/(a + b)$. For $c \neq -1$ its discriminant is

$$(a + b)^2 + 4c(1 + c)(b - a)^2 = (1 + 2c)^2(a - b)^2 + 4ab > 0.$$

The root where the polynomial crosses from negative to positive is exactly the plus-square-root expression in (4), for either sign of $1 + c$. This proves $M = L_c$ and also proves strict internality of L_c. Its symmetry, homogeneity and diagonal identity follow directly from the formula, or from uniqueness of the interior root.

For completeness we prove symmetry coincidence on this entire real parameter range. Fix $a < b$, put $m = L_c(a, b)$, $u = N(a, b)$, and let v be (1). The positive odds relation

$$\frac{v-a}{b-v} = \left(\frac{m-a}{b-m}\right)^2 \frac{b-u}{u-a}$$

shows $a < v < b$. It also shows that u and v lie on opposite sides of m, unless both equal m. Thus this pointwise construction defines a strict symmetric mean $V = S_{L_c}(N)$. If $D = (u-a)(m-b)^2 - (m-a)^2(u-b) > 0$ is its denominator, polynomial expansion gives the identity

$$D\, Q_c(u, v; m) = -(a-b)(u-m)^2 Q_c(a, b; m) = 0. \tag{8}$$

For $u \neq v$, m is between u and v and hence is the unique interior root, giving $L_c(u, v) = m$. For $u = v = m$ the same assertion is immediate. The diagonal $a = b$ is also immediate.

Finally, a value w satisfying $L_c(u, w) = m$ must satisfy the equation $Q_c(u, w; m) = 0$, which is linear in w. Its coefficient is $2cu - (1 + 2c)m$. This cannot vanish for a solvable equation: for $c = 0$ it equals $-m$; for $c \neq 0$, if it vanished the constant term would be $-m^2/(2c) \neq 0$. Therefore v is the unique functional-symmetry value. This proves the converse and finishes Theorem 1.

All asymptotic coefficients

Writing $L_c(x - t, x + t) = x + h$, (7) reduces to

$$xh + (1 + c)h^2 = ct^2.$$

With $H = h/x$ and $z = t^2/x^2$, this becomes $H + (1 + c)H^2 = cz$. The branch $H(0) = 0$ is analytic in z. Its coefficients $h = \sum_{n\geq 1} b_n t^{2n} x^{1-2n}$ satisfy

$$b_1 = c, \qquad b_n = -(1 + c) \sum_{j=1}^{n-1} b_j b_{n-j} \quad (n \geq 2).$$

The Catalan recurrence therefore proves

$$b_n = (-1)^{n-1} C_{n-1} c^n (1 + c)^{n-1}, \qquad C_k = \frac{1}{k+1}\binom{2k}{k}.$$

This includes $c = -1$ (the harmonic mean) and $c = 0$ (the arithmetic mean), where the higher terms vanish. Since $M = L_c$ globally, its two-variable expansion also agrees after replacing x by $x + (s+t)/2$ and the half-difference by $(t - s)/2$.

Four follow-up tasks

These are proposed questions, not additional solved credits.

Task 2. Classify all continuous strict symmetric homogeneous means whose two symmetries coincide for every strict symmetric mean, without assuming any diagonal asymptotic expansion.

Task 3. Classify all C^2 strict symmetric means on $(0, \infty)^2$ whose two symmetries coincide for every strict symmetric mean, without assuming homogeneity.

Task 4. Classify the analytic strict symmetric homogeneous means M satisfying only $M(A, S_M(A)) = M$, where A is the arithmetic mean. Determine whether this single test forces $M = L_c$.

Task 5. For fixed $R > 1$, $K > 0$ and $0 < \eta < 1/2$, determine the optimal modulus $\omega(\varepsilon) \to 0$, if one exists, bounding $\inf_c \sup_{[1,R]^2} |M - L_c|$ for C^4 symmetric homogeneous means with $\|M\|_{C^4([1,R]^2)} \leq K$, with $a+\eta(b-a) \leq M(a,b) \leq b-\eta(b-a)$ for $1 \leq a < b \leq R$, and with $|M(u, S_m(u)) - m| \leq \varepsilon$ for all such a, b and $a < u < b$. Here $m = M(a,b)$ and $S_m(u)$ is (1).

Verification and review

The proof is analytic and algebraic. The accompanying exact SymPy verifier checks the rational identities, quadratic endpoint signs, discriminant, local second-order coefficient and Catalan recurrence. These checks supplement the proof; they do not replace the limiting argument or the universal quantifiers. The current 16-page manuscript and 15-page final article were read completely. Source discrepancies and the limits of the prior-claim search are in the literature audit. This is author-agent review, without independent peer review.

References

[1] L. Mihoković, *Coinciding Mean of the Two Symmetries on the Set of Mean Functions*, Axioms **12**(3) (2023), 238, doi:10.3390/axioms12030238. Current manuscript: arXiv:2112.07489v3.